

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering An Autonomous Institute (Permanently affiliated to Savitribai Phule Pune University)	SEMESTER- VI
ASSIGNMENT- 5		
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1. Objective

The objective of this assignment is to develop a simple functional web application using Node.js and Express. The application simulates a travel agency website where users can view available travel packages and make bookings. MongoDB is used as a NoSQL database to store and manage data dynamically.

2. Learning Outcomes

After completing this assignment, I was able to:

- Understand backend development using Node.js and Express.js
- Perform CRUD operations using MongoDB
- Design RESTful APIs for handling requests
- Connect frontend with backend using fetch API
- Handle form submission and dynamic data rendering
- Understand client-server architecture

3. Brief Theory

Node.js is a JavaScript runtime environment that allows execution of JavaScript on the server side.

Express.js is a lightweight web framework used to build web applications and APIs efficiently.

MongoDB is a NoSQL database that stores data in the form of documents (JSON-like format), making it

flexible and easy to use with JavaScript-based applications.

In this project:

- The frontend is built using HTML, CSS, and JavaScript •

The backend is developed using Node.js and Express

- MongoDB is used to store travel packages and booking data

The application follows a **client-server architecture**, where: • The

client sends HTTP requests

- The server processes the request
- The database stores/retrieves data
- The response is sent back to the client

4. System Architecture (Brief)

The system consists of three main components:

- **Frontend:** User interface for viewing packages and booking • **Backend**

(Express Server): Handles API requests and business logic • **Database**

(MongoDB): Stores packages and booking details Flow:

1. User interacts with frontend
2. Request sent to server (API)
3. Server processes request
4. Data stored/retrieved from database
5. Response displayed on UI

5. Features Implemented

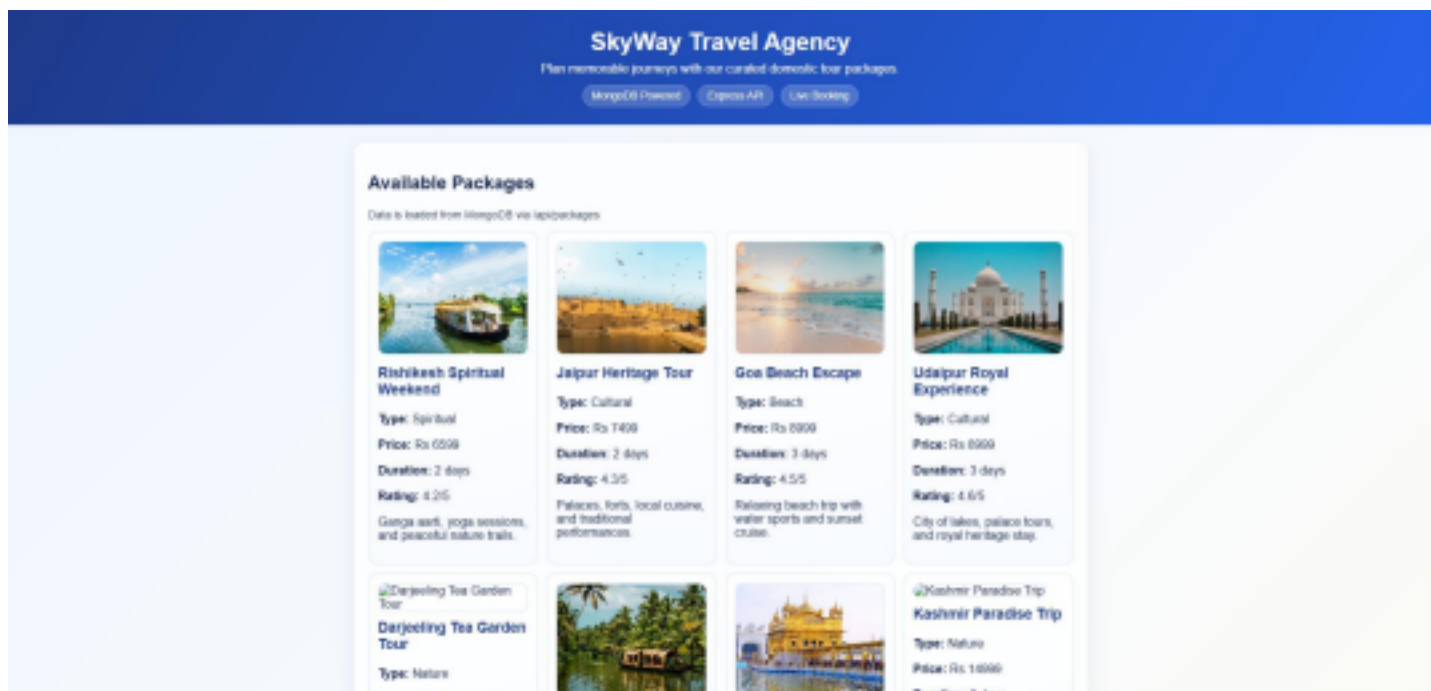
- Display travel packages dynamically
- Add new booking using form submission
- Store booking data in MongoDB

- Fetch and display bookings
- REST API implementation (GET, POST)
- Dynamic UI updates using JavaScript
- Responsive design

6. Program Code and Output

Program Code: [Click here](#)

Output Screenshot:



Book Your Trip

Name

Email

Select Package
-- Choose a package --

No. of Travelers
1

[Confirm Booking](#)

Recent Bookings (From Database)

[Refresh Bookings](#)

- Sonam booked Island Adventure Camp for 3 traveler(s) - Rs 32487
- Adarsh booked Kerala Backwater Retreat for 2 traveler(s) - Rs 21998
- Aditya and Nish booked Goa Beach Escape for 4 traveler(s) - Rs 35898

SkyWay Travel Agency
Assignment done: Node.js + Express, Post, MongooseDB, Sails
+ MongoDB + Mongoose

Office
Pune, Maharashtra, India

Contact
Email: help@skywaytravels.com

Hours
Mon-Sat, 9:00 AM - 7:30 PM

7. Conclusion

In this assignment, I developed a functional travel agency website using Node.js, Express, and MongoDB. I learned how to build a backend server, create APIs, and connect it with a frontend interface. The project helped me understand real-world web application development and improved my full-stack development skills.

